Results: Curriculum Graph and Assessment Audits

Graph validity, prerequisite depth, and assessment-quality gates I

The assembled curriculum is a valid directed acyclic graph of **630 skills** in **111 subjects** joined by **1199 prerequisite edges**, spanning all 8 strata with a maximum prerequisite depth of 75 and two foundational roots (set theory and the probability axioms). Validation passes: no cycles, no dangling prerequisites, and every applications skill is reachable from a foundational root. fig. 1 lays the graph out by stratum and depth, uses wrapped stratum-count labels and root annotations, and reports the maximum-depth summary.

The learner-quality gate then checks whether that valid graph is also usable as a micro-learning catalog. Each skill has at least 4 failable assessment items including a transfer-oriented multiple-choice check. Answer position and answer length are then audited per answer-count bucket, in every bucket carrying enough single-choice items to read. Each option slot has to be correct at a rate inside a band around chance, between 0.10 and 0.45 of the bucket, so no slot either dominates or sits unused. The length check is two-sided: the rate at which the correct option is the strictly longest answer and the rate at which it is the strictly shortest answer must both land between 0.12 and 0.42 of the bucket. Bounding only the long end would relocate the cue rather than remove it, because an answer that is never the longest teaches a learner to pick the shortest one. Multiple-choice

correctness is kept from collapsing onto a template: no option index sits in the correct set for more than 0.60 of multiple-choice items, and no single correct subset covers more than 0.30 of them. Objective strings may repeat, but only within a bound — at most four skills may share one objective before the gate reports templated filler. No single name-normalized answer body is reused across more than 0.12 of skills, with the declared shared rubric exempted as described below. Skill tags include both stratum and topic context, and backward cross-stratum references are kept out of the exported prerequisite path.

The one deliberately shared instrument is disclosed rather than hidden. Each skill carries bespoke domain assessment items; most skills additionally carry a shared transfer-and-misuse rubric, a uniform metacognitive instrument applied across the curriculum and marked as such. The answer-reuse gate exempts that declared rubric, requires every skill to retain at least one bespoke item, and bounds every other name-normalized answer body to at most 0.12 of skills. The exemption is what makes that bound readable rather than misleading: each per-family review rubric spans a whole stratum-by-role family, a footprint above a tenth of the curriculum and therefore above the bound the gate enforces on authored answers. We name the instrument and the exemption instead of letting a shared rubric sit quietly under a threshold, and the bespoke-item requirement keeps the rubric scaffolding rather than a substitute for per-skill assessment.

These checks establish structural assessment discipline, not psychometric validity. They show that quiz-gated completion is not thin, duplicated, or position-biased under the local validators; they do not estimate item reliability, construct validity, learner ability, retention, transfer, or differential item functioning.

The headline depth of 75 is not a single conceptual chain. It is dominated by the per-stratum mastery and open-science review ladders, which sequence skills for completeness rather than by genuine conceptual prerequisite. Excluding only those review and meta scaffold families and keeping every genuine concept family (mathematics through applications, plus the dynamical, geometric, hierarchical, implementation and cognitive-security families), the substantive concept chain a learner must climb is 33 deep; the difference is review and mastery sequencing, not conceptual dependency. We report both so the headline number is not

read as the depth of the conceptual curriculum.

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Figure 1: The Active Inference curriculum as a prerequisite DAG generated from the current curriculum graph after structural checks. Each point is one skill, color encodes the pedagogical stratum, horizontal position follows the stratum order, vertical position follows longest prerequisite depth, and gray segments encode exported learning-path dependencies. Wrapped stratum chips report live skill totals, the two foundational roots are labeled, and the inset reports the live skill, subject, edge, and maximum-depth counts. This figure is an overview for structure and reachability; it does not by itself certify prose quality, assessment quality, or scholarship support.

The full DAG is the most faithful view, but it is intentionally dense: learners need to see both the individual prerequisite chains and the curriculum's aggregate traffic pattern. fig. 2 adds that second view. Rows are prerequisite strata, columns are dependent strata, and the right-hand bar gives the node count in each stratum. Diagonal cells show within-stratum reinforcement; upper-triangular cells show forward conceptual transitions. A populated lower-triangular cell would mean a backward learning-path dependency, which the quality gate rejects before export.

Graph validity, prerequisite depth, and assessment-quality gates I

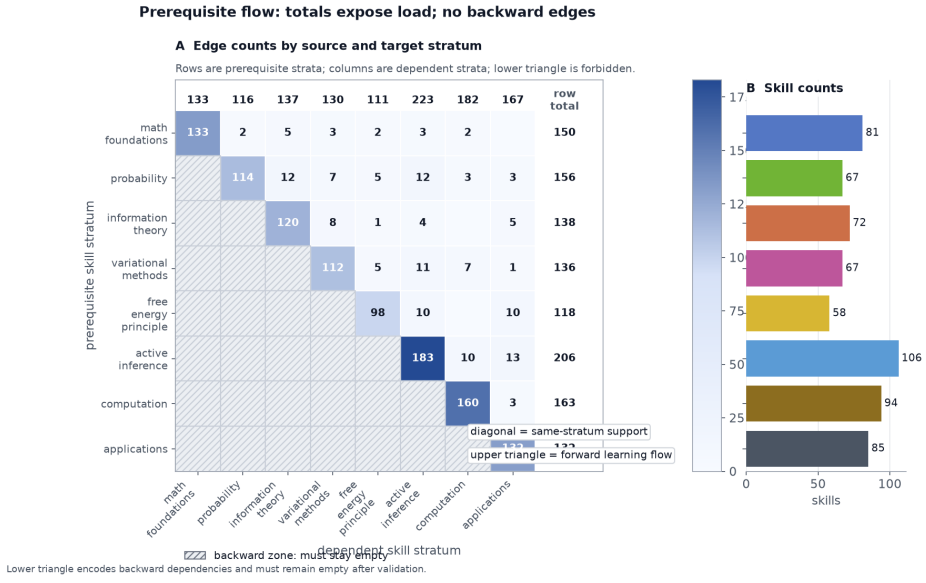


Figure 2: Prerequisite-edge flow across strata generated from every exported learning-path edge. Rows are prerequisite strata, columns are dependent strata, cell labels count edges, and the side bar reports how many skills live in each stratum. Diagonal cells show within-stratum reinforcement, upper-triangular cells show forward conceptual transitions, and a populated lower-triangular cell would expose a backward prerequisite that should be rejected before export.

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The same graph also needs diagnostic views for authors. fig. 3 orders high-outdegree skills by how much downstream curriculum they unlock, so review can focus on nodes where a weak explanation would create wide learner risk. fig. 4 complements that global view with a representative long dependency path. The figure keeps the route line readable while a checkpoint ledger names the start, stratum transitions, and endpoint, so a reader can inspect one concrete route without zooming the full DAG or relying on angled micro-labels.

Graph validity, prerequisite depth, and assessment-quality gates I

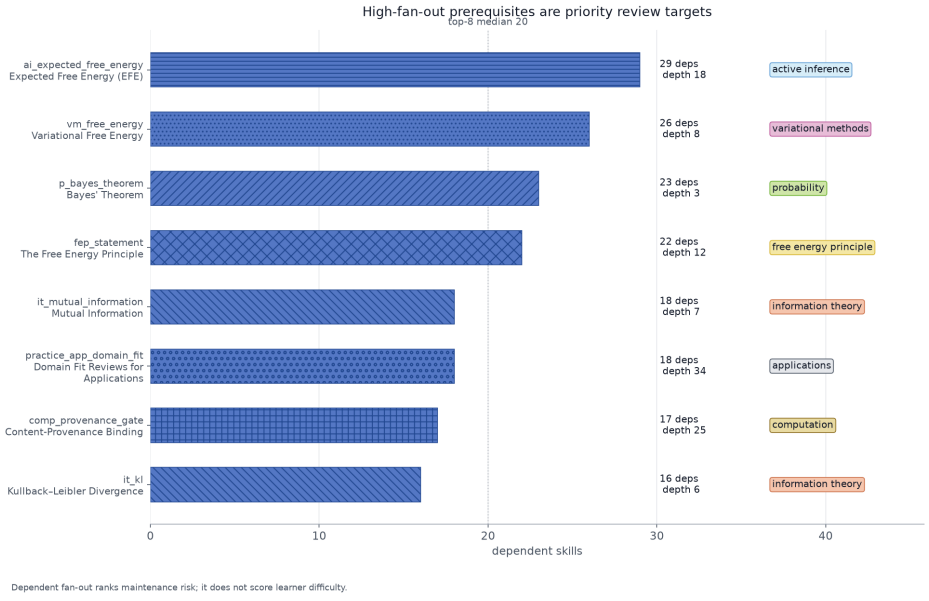


Figure 3: Graph bottlenecks in the exported prerequisite network generated from downstream dependent counts. Bars rank the top fan-out skills, y-axis labels pair skill ids with readable names, hatching and stratum chips identify pedagogical layer, and annotations report dependent count and prerequisite depth for each high-impact node. The figure is a review-priority surface: it identifies where a weak explanation, stale citation, or thin quiz would propagate to many later skills even though the graph remains valid.

Graph validity, prerequisite depth, and assessment-quality gates II

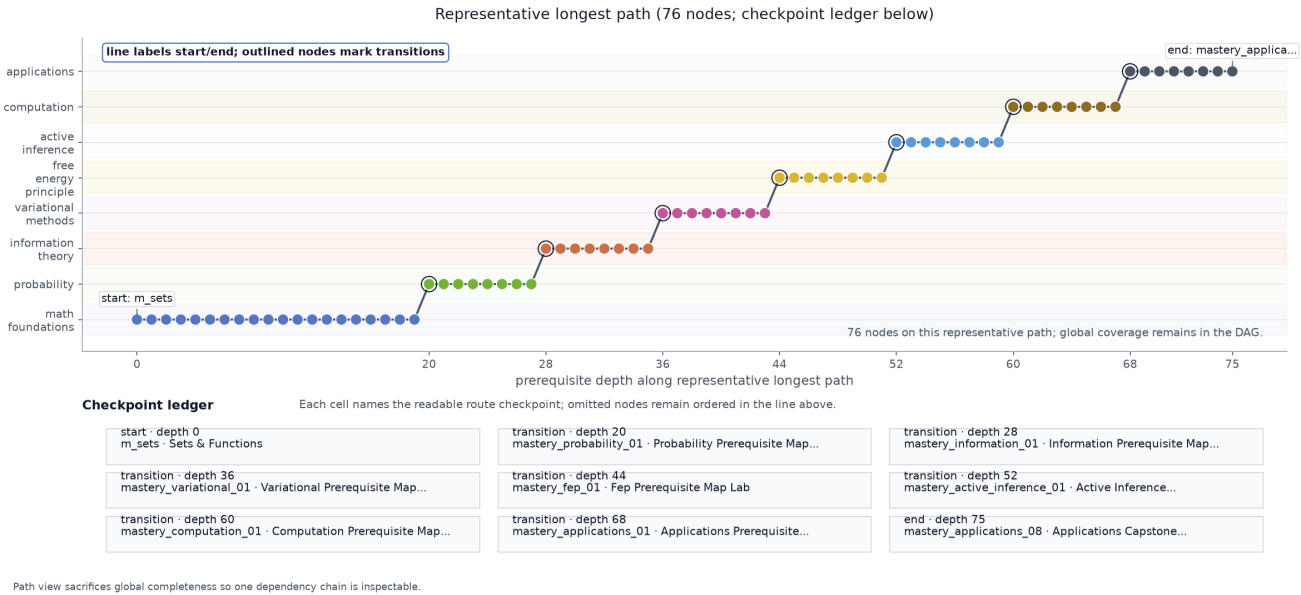


Figure 4: Representative skill-neighborhood path generated from the longest-path diagnostic in the current graph. The line orders skills by prerequisite depth, stratum bands and vertical position encode pedagogical layer, and a checkpoint ledger names the start, every stratum transition, and the endpoint. This view deliberately trades global completeness for readability: it lets authors audit one concrete foundation-to-application pathway without relying on the dense full-DAG overview or tiny rotated labels.

fig. 5 audits the semantic-prerequisite gate, which checks that no skill is assessed on one of the gate's tracked load-bearing concepts unless a skill defining that concept sits in its transitive prerequisite closure. The gate reads assessment text only, and it considers only definers at or before the using skill's own stratum, so a forward reference to a later concept is not counted as a missing prerequisite. For each concept the figure reports how many skills reach the defining skill through their prerequisite closure and how many are assessed on the concept, so an inversion in which a learner met a concept before they could have learned it would be visible rather than buried in the dense graph.

Graph validity, prerequisite depth, and assessment-quality gates I

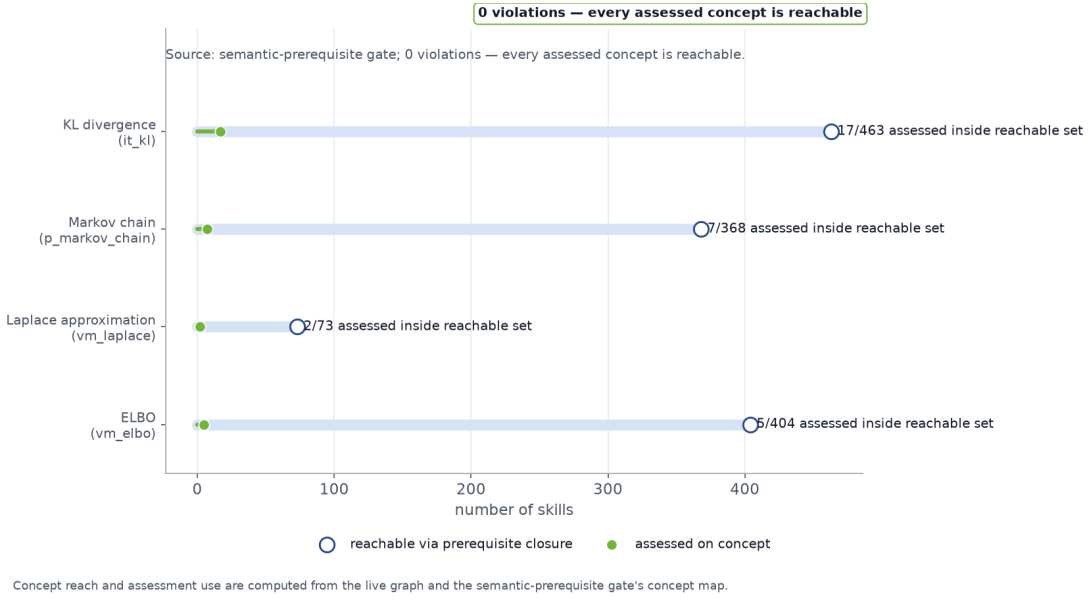


Figure 5: Semantic-prerequisite coverage generated from the live graph and the quality gate's concept map. For each load-bearing concept the open marker and stem report skills whose prerequisite closure reaches the concept's defining skill, while the filled marker and direct label report the assessed subset inside that reachable set. The subtitle and badge report the gate verdict so a passing curriculum shows zero skills tested on a concept they could not yet have learned.

The quiz surface is itself part of the curriculum result rather than a decorative add-on. fig. 6 summarizes the assessment-depth invariant by stratum. Every skill now carries the same 4-question floor, so completion is not certified by a single recognition item or by a thin two-question check.

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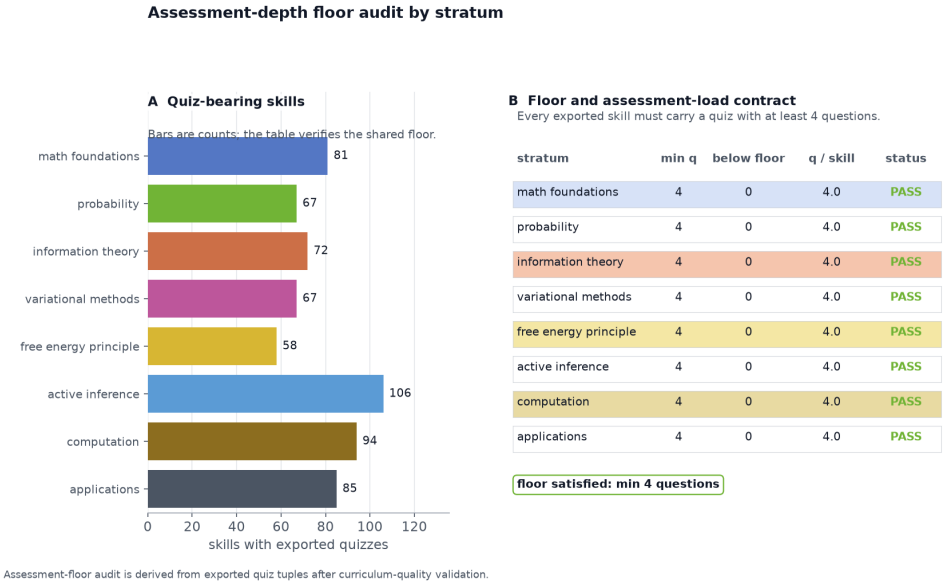


Figure 6: Quiz assessment depth by stratum generated from the skill assessment tuples. The bar panel counts quiz-bearing skills per pedagogical stratum, and the accompanying audit table reports, per stratum, the minimum questions per skill, below-floor counts, questions per skill, and a PASS/REVIEW verdict against the configured 4-question failable floor. The intended check is not average quiz length but minimum learner friction: any below-floor mass would show that a skill can be completed without the configured 4-question floor.

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The latest expansion also needs a balance check. fig. 7 summarizes skills, subjects, assessment questions and incoming prerequisite edges by stratum. This makes a different false-certification path visible: the graph can be larger while still overloading one pedagogical layer or leaving another underdeveloped.

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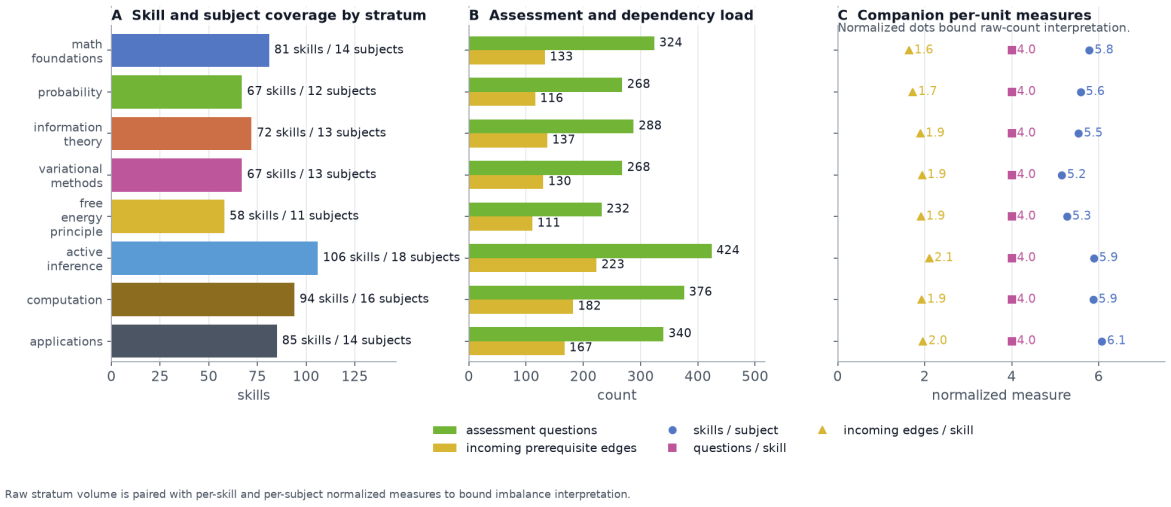


Figure 7: Curriculum stratum-balance audit generated from current graph statistics after structural checks. Three panels report the same underlying build: the left panel shows skills and subjects per pedagogical stratum, the middle panel shows assessment-question volume and incoming prerequisite edges, and the right panel shows companion per-unit (per-skill and per-subject) normalized measures. Reading the panels together shows whether expansion is distributed across the learning path or whether one layer is absorbing disproportionate topic, quiz, or dependency load. The figure bounds the claim to catalog structure; it does not measure learner outcomes.

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Having established the graph surface, the next result is source support. These audits ask whether the manuscript's scholarly claims are connected to the roles its citations are supposed to play.